

Everyone Was Right: Lexical Representations in Language Models Are Both Abstract and Holistic

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Abstract

Lexical representations are central to linguistic theory, yet to what extent they’re memorized or abstract remains a core debate. In the present study, we investigate how LLMs navigate this by decomposing contextual word representations into three additive components: an item component, a class component, and an idiosyncratic word-class interaction. After validating our measurement on an artificial language, we evaluate six pretrained transformers across two linguistic categories: part of speech and grammatical role. We find that the idiosyncratic interaction term is consistently positive and grows substantially across layers. Crucially, while item and class axes remain nearly orthogonal, the interaction term systematically overlaps with both. Together, our results demonstrate that LLMs do not commit exclusively to pure abstraction or exemplar storage; rather, they construct representations that are simultaneously both abstract and holistic.

1 Introduction

Lexical representations are deceptively complex: a single word can carry vastly different meanings depending on the context, yet these meanings remain recognizably related. The categorization literature has long recognized this tension, noting that assigning a meaning to a specific category is rarely straightforward. Simple rule-based approaches quickly break down: a three-legged cat is undeniably a cat, even if it violates one of the criteria used to define the category. Ultimately, defining a lexical category in precise terms may be impossible. Yet, as U.S. Supreme Court Justice Potter Stewart famously observed, we may not be able to define a category precisely, but we nevertheless “know it when [we] see it.”¹

¹In *Jacobellis v. Ohio* (1964), Justice Stewart famously addressed the difficulty of defining obscenity, remarking that while he may not be able to formulate a precise definition, he could still recognize obscene material when he saw it.

This central tradeoff, that representations are simultaneously context-specific yet broadly generalizable, has fueled long-standing debates over the nature of lexical representations (Kapatsinski, 2010; Pierrehumbert, 2016). The two leading frameworks offer distinct solutions: abstraction theories posit that humans construct generalized mental representations stripped of specific context, while exemplar theories argue that representations are built from distinct, context-rich memory traces.

In the present study, we investigate how Large Language Models (LLMs) navigate this tradeoff. Specifically, we examine the extent to which LLM lexical representations comprise abstract, generalizable features versus context-specific exemplars.

1.1 Evidence for Abstractions

Abstraction theories offer an intuitive account of the context invariance of language. At every level of language, there is substantial variability across contexts, from phonetic differences between speakers, to syntactic constructions that can take on a variety of lexical items, to the same word taking on different meanings in different contexts. The existence of abstractions accounts for the ability of humans to generalize across these contexts and recognize the same underlying representation despite this variability. Indeed, researchers have throughout the years compiled an abundance of evidence for abstractions.

Productivity is the most direct of these. Berko (1958) showed that young children can produce the correct plural of a word they have never encountered before, such as *wug*. Since that plural was never encountered before, it could not possibly have been stored in memory and it instead must have been built from an abstract representation of the plural. The same conclusion has been reached from neural evidence. Using MEG, Stockall and Marantz (2006) found that irregular forms such as *gave* prime their root as strongly as regular forms

do, while pairs that merely look and sound similar without being morphologically related don't prime them at all. Because the priming tracks morphological structure rather than surface similarity, even the representations of irregular forms appear to have some abstract component to them.

Even something as deceptively simple as word order preferences, such as binomial ordering preferences, is shaped by abstract knowledge (Morgan and Levy, 2016). For example, humans have strong preferences for the order of the two words in coordinate phrases, or binomials, such as *bread and butter* over *butter and bread*. Those preferences are to some extent driven by abstract preferences such as a preference for male-coded words before female-coded words. Crucially, speakers' preferences for binomials that they have never encountered before are driven by these abstract properties, so the preferences cannot be stored with any particular pair of words.

1.2 Evidence for Exemplars

Exemplar theories, in contrast, argue that abstract representations are not necessary. Instead, linguistic knowledge may be stored as context-rich memory traces whose properties vary with experience. Much of the evidence for this comes from the phonetics literature, showing that phonetic representations are surprisingly rich, as well as the psycholinguistic literature's line of research demonstrating the holistic storage of multi-morphemic multi-word phrases (Bybee, 1999; Houghton and Morgan, 2023; Houghton et al., 2025, 2026).

The strongest evidence for exemplars comes from frequency in favorable context effects (FFC, Kapatsinski, 2026; Raymond and Brown, 2012). FFC effects describe a phenomenon in which a form's behavior is predicted by how often it has occurred in a context that conditions a change rather than by how often it has occurred overall (Kapatsinski, 2026). FFC effects show that it is not raw frequency, but rather frequency in a specific context, that is stored. For example, Raymond et al. (2016) found that English words ending in /t/ or /d/ were more likely to be deleted if they were words that frequently occurred before a consonant (the conditioning context for that deletion), regardless of the words' overall frequency. Crucially, this deletion was observed even when the word occurred outside of a context that favored the conditioning (Bybee, 2002). Context-specific storage is also evidenced by phonetic reduction (Bybee

and Scheibman, 1999), gradient phonetic representations (McMurray et al., 2008; Miller, 1994; Palmeri et al., 1993), reading times for binomials (Sivanova-Chanturia et al., 2011), word-order preferences (Morgan and Levy, 2015, 2016, 2024), and recognition times (Houghton, 2025; Kapatsinski and Radicke, 2009).

The same question arises within individual words, where debates have long taken place over whether inflected forms are retrieved whole or assembled from a stem and an affix (Stemberger and MacWhinney, 2004; Stemberger and MacWhinney, 1986; see also Schreuder and Baayen, 1997). Beyond morphologically complex words, a parallel question remains at the morphologically simple level. Consider *walk*, which can function as both a noun and a verb: if its representation as a verb were fully determined by a representation of *walk* and a representation of the verb category, then nothing in it would be specific to the *walk-verb* combination, whereas if some part were unique to *walk* as a verb, the representation would contain information that cannot be reduced to its constituent parts. Word embeddings offer a way of asking this directly, since they give a separate representation of a word in each category it occurs in. Recent work in this vein shows that abstractions can be narrower than might be expected (Nikolaev et al., 2025; Shafaei-Bajestan et al., 2024), but such narrowness still reflects information shared across words within a more specific class. What remains unclear is whether representations also contain information specific to an individual word-category combination.

1.3 Present study

The evidence for both abstraction and item-specificity therefore raises a broader question about the structure of linguistic representations: is a representation of a specific linguistic construction a combination of abstract features, or does that representation also contain item-specific knowledge that is irreducible to these component parts? This is difficult to settle behaviorally, since a learner with no abstract representations at all can look like one that abstracts (Houghton and Kapatsinski, 2026), but it is a property of a representation, and unlike in humans, model representations can be examined directly (Elhage et al., 2022; Hinton et al., 1986; Pluth et al., 2026).

In the present study we examine whether linguistic representations contain both abstract and

item-specific information by decomposing a word-in-class representation into three components: a class component (e.g., a verb component), and item component (e.g., *walk* regardless of the part of speech), and an item-of-a-specific-class component (e.g., *walk* as a verb). We first validate this decomposition on a model trained on an artificial language, then apply it to six pretrained language models across two linguistic constructions.

Our main contributions are as follows:

- We introduce a method for decomposing word-in-context representations into lexical, grammatical, and word-class components, and validate it on an artificial language.
- Across six pretrained models, the word-class component is consistently present and grows with depth; in the artificial language its size is governed more by representational capacity than by the language’s composition. The lexical and grammatical components are nearly orthogonal, while the word-class component overlaps with both.
- We show that the word-class component contains information about the class itself beyond the shared class representation.

2 Experiment 1: Artificial Languages

In Experiment 1 we test our measurement on an artificial language whose features we control. This allows us to validate our measurement while simultaneously examining how different features of the language affect a model’s representations.

2.1 Dataset

In order to examine the separability of linguistic representations, we constructed an artificial language which comprises a probability distribution over a vocabulary of 600 tokens. Every observation is a combination of a **form** (one of 60), a **class** (one of 4), and a **context** (drawn from a pool of 400). The motivation for this setup is to give each word a naturalistic environment: each word occurs in a context that favors a specific class-membership (e.g., *walk* as a verb occurs in a different context than *walk* as a noun). Each form is observed 24 times in each class, each time in a different context, mirroring the patterns of human language. Consider *walk* used as a verb in one of those contexts. Four things determine the distribution over the next token. *Walk* itself contributes the same amount

regardless of the class it is used in, and the verb class also contributes the same amount for every item. The third contribution is what is specific to *walk*-as-a-verb over and above those two: every form and every class has its own random vector, and this term multiplies *walk*’s by the verb’s, entry by entry. The context contributes the rest, which is noise as far as our question goes, though the model still has to encode it in order to accurately learn the language. The four are weighted, summed and passed through a softmax layer in order to create the ground truth probability distribution over next tokens for a given input:

$$P(\text{next token} \mid f, c, x) = \text{softmax} \left(s \left[w_{\alpha} \text{item}(f) + w_{\beta} \text{class}(c) + w_{\gamma} \text{int}(f, c) + w_{\text{ctx}} \text{context}(x) \right] \right) \quad (1)$$

Each term is isotropically scaled to unit variance before being weighted,² so changing the weights changes the mix of the three effects without changing how much signal there is in total.

The weights w_{α} , w_{β} , and w_{γ} in Equation 1 control how much of the true distribution each of the three effects contributes, and we cross them factorially over $\{0, 0.5, 1, 2\}$: zero removes an effect from the language altogether, and the other three values make an effect half as strong as another, equal to it, or twice it. Similarly, w_{ctx} governs how much the context contributes, so we cross those languages with three context weights: $\{0.5, 2.0, 6.0\}$, ranging from context being a minor part of the true distribution to the dominant one. The four weights are subsequently rescaled together to unit length, such that $w_{\alpha}^2 + w_{\beta}^2 + w_{\gamma}^2 + w_{\text{ctx}}^2 = 1$.³ The ranks of the item and class are held fixed at 8 and 4 respectively but each language is crossed with four interaction effect ranks: $\{4, 16, 64, 177\}$. Additionally, we train a model on each of the resulting languages at hidden widths of 8, 16, 32, 128, 256, with either a linear or a ReLU activation. The whole grid is then run with five random seeds giving 37,800

²Specifically, each observation consists of a form in a specific class and a context, paired with that term’s contribution to the next-token distribution for that observation. We normalize this dataset by dividing each value by the total standard deviation across all observations.

³This makes the resulting languages more comparable by enforcing the same total signal (i.e., the variance of the weighted sum in Equation 1 is held constant), despite individual differences in the amount that each individual term contributes.

runs in all, of which 99.6% trained to convergence; Appendix A lists every factor and its levels.

2.2 Method

2.2.1 The learner

The learner model is a multi-layer perceptron (MLP) with a single hidden layer of width d and either a linear or a ReLU activation function.⁴ The model takes as its input three pieces of information: form, class, and context. Each has its own embedding, and the three are concatenated and passed through the model which then outputs a probability distribution over the vocabulary (Figure 1). The training objective is thus to minimize cross-entropy between the model’s predicted distribution over the vocabulary and the ground-truth distribution.

2.2.2 Decomposing a representation

In order to measure the item-specific part of the representation, we decompose a word-in-context representation into a lexical component, a class component, and an idiosyncratic component. In Experiment 1, the representation we decompose is the MLP’s hidden layer representation. Let $M[i, c] \in \mathbb{R}^d$ be the mean of those hidden states for item i used at class c (Appendix B). We decompose those item-by-class means by centering, so that:

$$M[i, c] = \mu + \alpha(i) + \beta(c) + \gamma(i, c), \quad (2)$$

where μ is the grand mean; $\alpha(i)$ is item i ’s mean across classes minus μ which conceptually is the representation unique to the item; $\beta(c)$ is the mean across items at class c minus μ which conceptually is the representation unique to the class (e.g., noun vs verb); and $\gamma(i, c)$ is what item i at class c holds that neither marginal accounts for which conceptually is the item-specific representation in a specific class (Figure 2). Each representation for a given item in a given class is exactly the sum of the four terms of Equation 2 and no two of them vary together across those representations, so the squared lengths of each term (excluding μ) sum to the total and we report each as a share.

Each of these averages is taken over a sample of observations that differ only in context, so each carries some of the context term of Equation 1 as noise. We remove it by dividing the observations

into two evenly sized splits, running the decomposition on each split separately, and estimating each component’s size as the dot product of its two split estimates, which is large when the splits agree about the component and near zero when they do not. As a result, noise that differs between the splits is independent and cancels.

For each representation we report two measurements. The first measure is the fraction of the representation accounted for by lexical, class, and idiosyncratic information: each component’s squared length, summed over its vectors and divided by the total, so that the three fractions sum to one. The second measure quantifies the extent to which those components occupy the same geometric directions. Specifically, we project the item vectors onto the class direction and record the fraction of their squared length that survives, and we project the interaction onto the smallest subspace containing both the item and the class vectors (i.e., the span of the two subspaces).⁵

To quantify whether a size is meaningful, we redraw the split two hundred times and estimate how often its confidence interval excludes zero; for components whose value is exactly zero this flags an effect about 5% of the time (Appendix C), the false-positive rate expected of a 95% confidence interval.

To quantify whether an overlap is meaningful, we project the effect, held exactly as measured, onto two hundred random subspaces of the same rank, which gives the distribution of overlaps an unrelated subspace would yield. An overlap below that baseline indicates meaningful orthogonality; an overlap above the baseline indicates meaningful non-orthogonality. We report each overlap as a multiple of the baseline’s median, with how many measurements fall outside its 95% interval.

It’s important to note that the weights in Equation 1 describe the ground-truth distribution the model is trained to match, while the measure decomposes the representation that the model *learns*, so the two sets of numbers are not necessarily analogous: a learner handed a language that is 85% interaction can learn a representation in which the interaction is far smaller (or larger). Appendix C reports a separate check in which components were

⁴All code and analyses in this paper can be found here: <https://anonymous.4open.science/r/representation-separability-D254/>.

⁵The class effect spans at most one direction fewer than the number of classes, so it is low-dimensional in both experiments: one direction in Experiment 2 and at most three in Experiment 1. Its projection into the item subspace, which has many more dimensions, has high overlap by chance; we therefore do not report it.

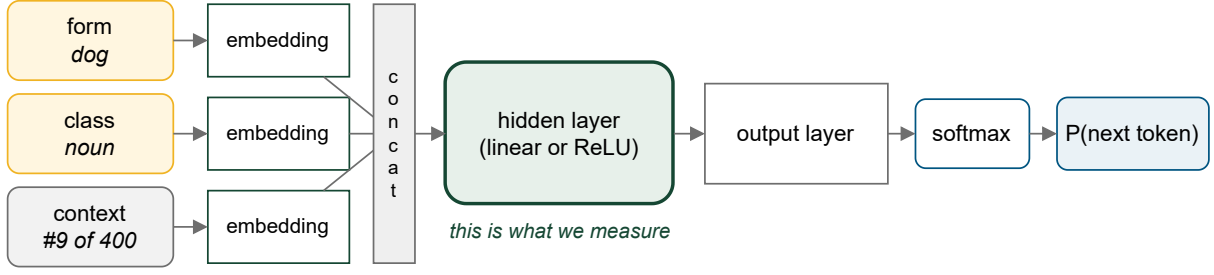


Figure 1: Structure of our model. Every form is observed 24 times at each class, each time in a different context; that variation across contexts is what the measurement later splits in two.

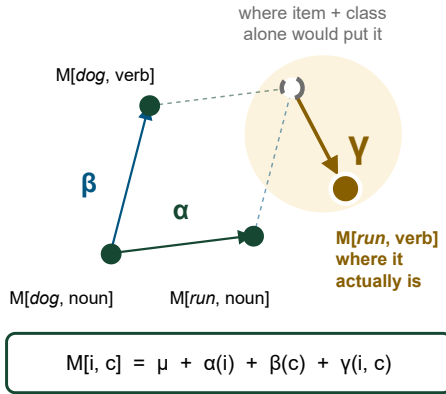


Figure 2: The four item-by-class means formed by two items in two classes. The arrow is $\gamma(i, c)$.

set directly in a representation and recovered almost exactly, which is what licenses interpreting the results below as facts about the learner rather than about the measure. We include runs only if at least one of the three components has a re-split interval excluding zero; where none does, the three shares divide noise.

2.2.3 Analysis

We model our data using an ordered beta regression (Kubinec, 2023) for the first measurement and a Gaussian model for the second measurement, both in *brms* (Bürkner, 2017) with default priors. For the first measurement, we modeled each component’s fraction of the representation as a function of its own effect size in the language, the hidden width, and the total rank of the planted structure (with interactions between each), with the other two components as controls, and random intercepts by language (language here refers to the 5 languages that share parameter settings but differ by seed). Equation 3 demonstrates the model for the item effect; the class and interaction models are analogous, but with their own effect being included as an interaction term rather than the item effect. All

predictors are centered. Coefficient values here indicate how much a specific parameter affects the amount that the model represents either item, class, or the interaction.

$$\begin{aligned} \text{ItemFraction} \sim & \text{ItemEffect} * \log_2(\text{Width}) \\ & * \log_2(\text{TotalRank}) \\ & + \text{ClassEffect} + \text{InteractionEffect} \\ & + (1 \mid \text{Language}) \end{aligned} \quad (3)$$

For our second measure, we modeled each overlap as a logged multiple of its own chance level with fixed-effects for each component, width, and total rank, with a random intercept for language (Equation 4). All predictors are centered for this model as well. Coefficient values here indicate the extent to which each parameter causes the item, class, and interaction dimensions to be orthogonal or non-orthogonal.

$$\begin{aligned} \log(\text{Overlap}/\text{Chance}) \sim & \text{ItemEffect} \\ & + \text{ClassEffect} + \text{InteractionEffect} \\ & + \log_2(\text{Width}) * \log_2(\text{TotalRank}) \\ & + (1 \mid \text{Language}) \end{aligned} \quad (4)$$

2.3 Results

Results are visualized in Figure 3, with the full coefficient results included in Appendix D.

Unsurprisingly, the effect size for each component increases as a function of the size of the effect planted in the language (and since the three components sum to 1, the other 2 components definitionally decrease). Of the three components, the item effect is strongest across every width (Figure 3).

The effect size of the interaction does not depend on its rank in the language ($\beta = -0.016$). Instead, it is the size of the hidden layer that determines how much of the interaction is encoded ($\beta = +0.568$):

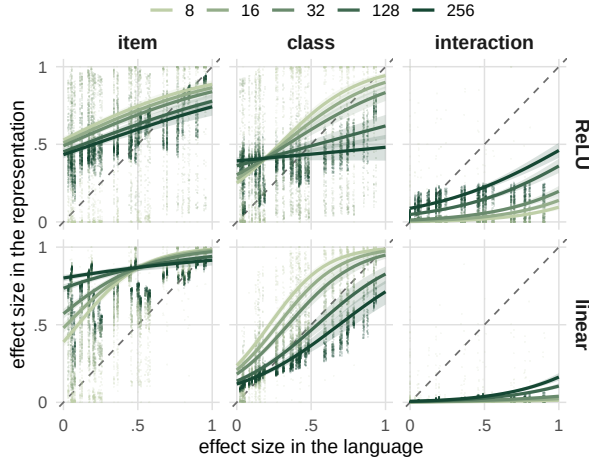


Figure 3: Ground-truth effect size of each component in the language (x) against its size in the MLP’s representations (y). Points are runs; lines are fitted values from Equation 3 with 95% intervals, coloured by hidden width. The dashed diagonal marks exact recovery.

the larger the hidden layer, the larger the size of the interaction it encodes. This is not the case for the item effect, which decreases as the hidden layer grows ($\beta = -0.095$), or for the class effect, which barely changes ($\beta = +0.030$). In the linear condition width is spent on the item effect at the class effect’s expense ($\beta = +0.247$ against $\beta = -0.233$); the interaction grows with width there too ($\beta = +0.187$) but stays small throughout.

The amount of overlap is also affected by the width of the hidden layer as well: For the ReLU models, the wider the hidden layer, the less the item and class effects share directions ($\beta = -0.398$). Similarly, the overlap between the interaction and the item-class subspace follows the same pattern ($\beta = -0.315$). The linear models show a different pattern, however. The item-class overlap shows no effect as a function of width, and the interaction effect’s overlap with the item-class space grows as a function of the width ($\beta = +0.101$), climbing from 0.46 to 0.78 times chance. It approaches chance without reaching it, and both overlaps stay below chance at every width under the linear learner (Appendix D). Smaller hidden layers thus do not merely represent less of the interaction, they also force the components onto shared directions.

3 Experiment 2: Language Models

Experiment 1 established that if the language a learner model encounters is not purely additive, the model will represent the language in a non-

separable manner, with the extent governed by the width of the model. Experiment 2 applies the same decomposition to the contextual representations of pretrained language models, across two linguistic categories: part of speech and grammatical role.

3.1 Dataset

In order to examine the separability of linguistic representations in models, we examined two linguistic categories: part of speech and grammatical role. For both, we concatenate six general-domain English Universal Dependencies treebanks (Nivre et al., 2020): EWT (Silveira et al., 2014), GUM (Zeldes, 2017), LinES (Ahrenberg, 2007), ParTUT (Sanguinetti and Bosco, 2015), GENTLE (Aoyama et al., 2023), and PUD (Zeman et al., 2017), giving 41,095 sentences and 706,323 word tokens. We measure two model families at three matched scales: OPT-BabyLM, trained on developmentally plausible quantities of child-available language (Houghton et al., 2026; Warstadt et al., 2023), and Pythia, trained on the Pile (Biderman et al., 2023), at 125M/160M, 350M/410M, and 1.3B/1.4B parameters, with widths of 768, 1024, and 2048. Appendix E gives parameter counts and training-token estimates.

3.2 Stimuli

Our stimuli comprise two linguistic categories: part of speech (NOUN versus VERB) and grammatical role (*nsubj* versus *obj*, restricted to nouns). For each category we select items that occur in both of its classes, such as *walk*, which occurs as both a noun and a verb. An item is a single inflected word form, not a lemma, so different inflections of one lemma are separate items: *walk* and *walking* are both items, each included only if it is itself attested in both classes. Pooling them under one lemma would collect *walk* and *walks* into a single item, and the two classes would then be read at different word forms, which the embedding layer could then separate before any contextualization has occurred. Capitalization is ignored when occurrences are grouped, but not in the text the models read, so a sentence-initial *Walk* counts toward the item *walk* and is tokenized as it appears there.⁶

An item is included only if it occurs at least ten times in each class. This yields 78 items for

⁶OPT-BabyLM’s tokenizer is uncased, so this choice has no effect for that family of models. On the other hand, Pythia’s tokenizer is not uncased; however, lowercasing the input would be the larger distortion, since a sentence-initial word is capitalized in written English which the model was trained on.

part of speech and 49 for grammatical role, with median counts per class running from 15 to 19 (Appendix F), inside the range we validated for the measurement (Appendix C).

3.3 Method

The decomposition, the two reported quantities, and the replication logic are identical to Experiment 1. For every token we take the hidden state of the word’s last subword at every layer of each model.⁷

Transformer hidden states contain a few rogue dimensions whose means and variance are both disproportionately high (Timkey and van Schijndel, 2021), and these would otherwise dominate any subspace estimated from the raw embeddings, so we divide each dimension by its standard deviation before decomposing (Appendix G).

3.4 Results

The results for the final layer are reported in Appendix H; the layer-by-layer results are shown in Figure 4.

The item effect is the largest component in both categories, running from 0.657 to 0.846 for part of speech and 0.828 to 0.904 for grammatical role. Similarly, the effect of class is mostly comparable (0.039 to 0.078 vs 0.050 to 0.098 for part of speech and grammatical role respectively). On the other hand, while the interaction effect is larger than the class effect for part of speech (0.096 to 0.245), the same is not true for grammatical role (0.045 to 0.094). This may indicate that part of speech is more lexicalized than grammatical role, perhaps unsurprisingly.

Another way to describe these results is with respect to the whole representation rather than the item-by-class structure: the item effect accounts for 19% to 38% of what varies in a token, the class effect 1% to 3%, and the interaction 2% to 8%, with the remaining 56% to 72% being the context a word appears in (See Appendix H).

Both the class effect and the interaction effect increase as a function of layer depth. The interaction is essentially absent at layer 0 in both categories, between 0.0009 to 0.0088 across every category and model, and rises to between 0.0455 to 0.2449 (Figure 4). This is not surprising since at the first layer, the model has yet to integrate enough con-

text to commit to the class of the representation. The increase in the effect across layer depth is also not an artifact of the learned absolute position embeddings: zeroing OPT-BabyLM’s leaves the class structure more word-specific rather than less, and Pythia, which encodes position with rotary embeddings inside attention rather than an additive input embedding, shows the interaction at least as strongly (Appendix I).

Turning to how those components are oriented, the class direction carries almost none of the item effect. At the deepest layer, item-class overlap runs 0.0035 to 0.0067 for part of speech and 0.0033 to 0.0062 for grammatical role, so projecting out the class direction leaves more than 99% of the item effect intact. While both overlaps are technically less orthogonal than a random subspace, the overlap itself is tiny, so the two are near-orthogonal rather than exactly orthogonal.

In contrast to the main effects, the interaction effect is not orthogonal to class or item. Across the six models, 0.106 to 0.239 of the interaction for part of speech is in the span of the two main effects, and for grammatical role 0.071 to 0.179 is, against nulls whose upper ends run 0.023 to 0.058 and 0.014 to 0.038 respectively. The smallest of these overlaps exceeds the largest of the nulls, so the pattern holds in every model. In other words, the small departure from orthogonality between item and class becomes a large one for the interaction.

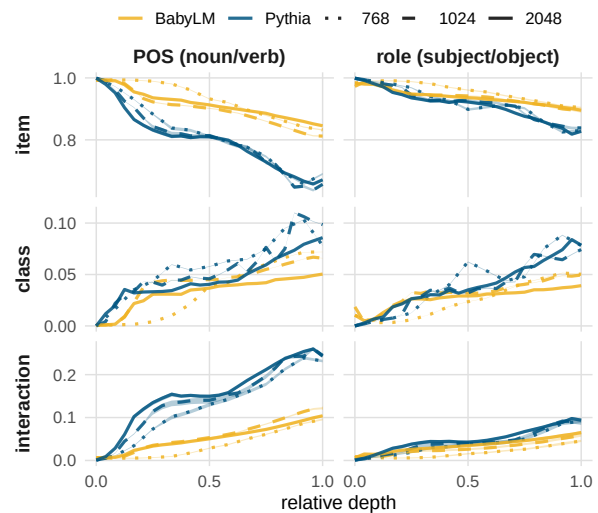


Figure 4: Fraction of the representation (ignoring context) accounted for by each component, summing to 1, against relative depth from the embedding layer (0) to the final layer (1). Each line is one model; colour is the training corpus, linetype the hidden width, shading the 95% interval.

⁷Almost every item is a single subword, so this choice has little bearing on the results: zero items are split into multiple subword tokens under Pythia’s tokenizer, and only eight are split under OPT-BabyLM’s tokenizer.

4 Experiment 3: Probing the Interaction

Experiment 2 shows that lexical representations contain an idiosyncratic component that can't be separated out into simple item and class representations, but neither shows what information is contained within the interaction. An item-by-class term captures any reliable difference between a word's uses in the two classes, and some of that difference may be correlated with the context, since subjects appear in different constructions from objects.

4.1 Method

We test whether the interaction encodes class information by probing its representation. If γ encodes the class, the class should be recoverable from the part of a token's representation that is neither the item effect nor the shared class direction, and we remove both explicitly. All probing happens *within* a single word, so we ask whether *walk*'s noun uses can be told from *walk*'s verb uses; the item effect is constant within a word and so cannot contribute. We then estimate β as the difference between the two class means pooled over all items, normalize it, and remove that component from every token, so that whatever separates the classes afterwards is not the direction they share.

What remains after removing those components is the interaction: the part of a word of a specific class that is neither a fact about the word nor a fact about the class. We fit a cross-validated logistic probe to it, five folds, with classes balanced by subsampling within each word so that chance is 0.50, and average the resulting accuracies over words. We include tokens only if they have at least ten occurrences in each class, the same threshold the stimuli are selected on. The whole procedure is then repeated with the class labels permuted within each word, which preserves every word's token count and class balance while destroying the labeling so that we have an appropriate control (He-witt and Liang, 2019). We run this at the deepest layer of the largest model in each family, for both categories.

4.2 Results

Results are reported in Table 1. For all models, the class is recoverable from the interaction at well above chance while the permuted null sits on chance, reaching 0.94 for part of speech and 0.84 for grammatical role. The null landing on chance

class	model	accuracy	null
part of speech	BabyLM 1.3B	0.90 [0.88, 0.91]	0.51 [0.49, 0.54]
part of speech	Pythia 1.4B	0.94 [0.92, 0.95]	0.50 [0.48, 0.53]
grammatical role	BabyLM 1.3B	0.80 [0.77, 0.83]	0.50 [0.47, 0.53]
grammatical role	Pythia 1.4B	0.84 [0.81, 0.86]	0.52 [0.49, 0.55]

Table 1: Accuracy of recovering the class from the interaction, at the deepest layer of the largest model in each family, compared to the null; brackets are bootstrapped intervals.

is what makes those accuracies interpretable, since it is the same probe, the same folds, and the same word-level averaging with only the labeling destroyed.

The results demonstrate that γ carries class information rather than generic correlated context. It does not show that γ is *only* the class label: context correlated with the class within a word survives both removals, so the interaction remains an upper bound.

5 Discussion and Conclusion

The present study examined whether a word's representation in a language model is exemplar-like or abstract and found that it contains elements of both. We decomposed a word's representation in a class into an item effect, a class effect, and a word-class interaction. The first two capture information shared across classes or words, respectively, while the interaction captures information specific to a particular word-class combination. Across six pretrained language models and two linguistic categories, this interaction was present in every model and category, nearly absent at the embedding layer, and grew throughout the network. The abstract components were nearly orthogonal, whereas the interaction overlapped with both, indicating that abstract and item-specific information are represented together rather than on separate axes.

Overall, the central result is not simply that language-model representations are sensitive to context, but that this context sensitivity is organized around particular word-class combinations. Language-model representations therefore combine abstraction with item-specific structure: they generalize across words and classes while retaining information about particular combinations.

Limitations

Our evaluation is restricted to English and relies on two linguistic categories tested across two model

families (OPT-BabyLM and Pythia) at three parameter scales. While these choices cover a range of architectures and linguistic phenomena, broader typological, categorical, and architectural validation remains a direction for future work.

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A The Artificial-Language Grid

Table 2 describes the language and the MLP parameters.

factor	levels
language	
item weight	0, 0.5, 1, 2
class weight	0, 0.5, 1, 2
interaction weight	0, 0.5, 1, 2
context weight	0.5, 2, 6
interaction rank	4, 16, 64, 177
learner	
hidden width	8, 16, 32, 128, 256
activation	linear, ReLU
replication	
seed	0, 1, 2, 3, 4

Table 2: The Experiment 1 language parameters. The first five rows define each artificial language, the next two the learner MLP model. Item and class ranks are held fixed so that the interaction’s rank is the only one that varies. Crossing every level gives 38,400 cells; the three languages in which all three effect weights are zero contain no item-by-class structure at all and are excluded, leaving the reported total.

Item rank is fixed at 8 and class rank at 4, so the structure a language requires is 12 directions plus the interaction’s rank, running from 16 to 189. Of the 37,800 runs, 99.6% trained to convergence.

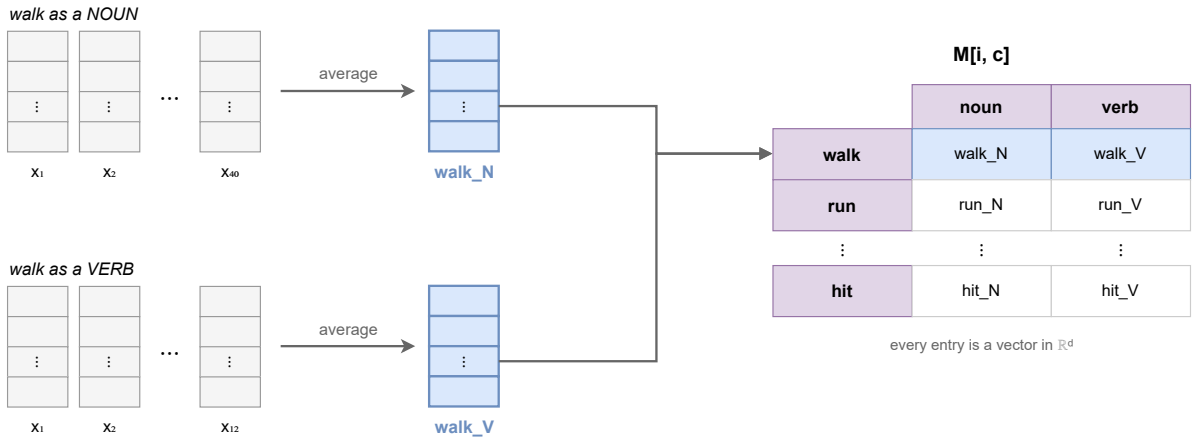


Figure 5: Each word in each class is observed in many contexts, and those observations are averaged to give that word's representation in that class. Every one of these averages is a vector in $\mathbb{R}^d$, and every item appears in every class.

A word used in a given class is observed in many contexts, and the representation differs a little in each. Averaging those observations gives that word's representation in that class: for items $i \in \{1, \dots, I\}$ crossed with classes $c \in \{1, \dots, C\}$, $M[i, c] \in \mathbb{R}^d$ is the mean representation of item i in class c (Figure 5). The three components are read off that grid (Figure 6).

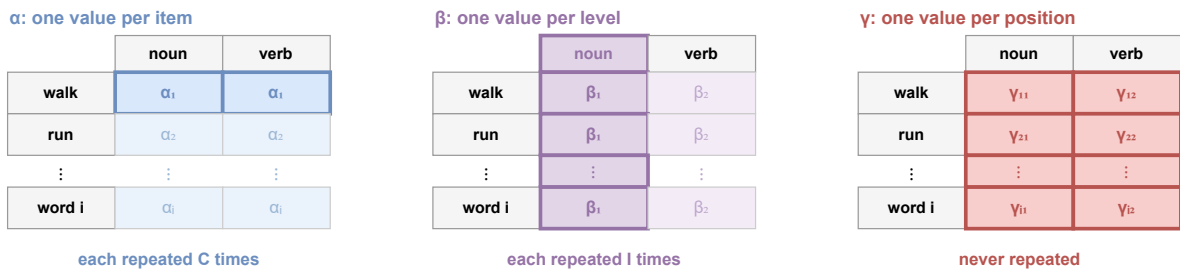


Figure 6: The three components across the grid of words and classes. An item effect is one value per word, the same in every class; a class effect is one value per class, the same for every word; and the interaction differs for every word in every class.

C Validating the Measure

Experiment 1 defines a ground truth language, however, it cannot distinguish a failure of the measure from the model representing the language differently than the structure it was given. Here we remove that ambiguity by constructing representations in which μ , α , β , and γ are set directly, with no model or training, and passing the result through the same analysis used for the language models. Ground truth therefore exists at the level where the measure operates.

Specifically, we generated representations with known values of μ , α , β , and γ , varying the number of items, classes, representation dimensions, observations per word and class, and noise level across the range used in the experiments. Components were centered to satisfy their defining constraints and rescaled to predetermined shares of the total squared length. The grid included the same factorial Experiment 1 run: the three components measured in Experiment 2, item counts from 27 to 180, two and four classes, widths from 8 to 2048, 10, 24, and 100 observations per item and class, three noise levels, and five seeds. We also generated orthogonal and known non-orthogonal item and class effects: the orthogonal condition draws α and β independently, while the non-orthogonal condition places a known fraction of β along α 's largest dimension. In total, this gave 252,000 runs.

At 24 observations per item and class, the measured interaction matches the one it was given across the whole range (Table 3), and is flagged in essentially every run where anything was present and about one in twenty where nothing was.

specified	runs	measured	flagged
0.000	6300	0.0000 [0.0000, 0.0094]	5%
0.030	900	0.0323 [0.0199, 0.0418]	100%
0.048	1800	0.0495 [0.0385, 0.0688]	100%
0.056	1800	0.0573 [0.0458, 0.0918]	100%
0.059	1800	0.0607 [0.0495, 0.1041]	100%
0.100	1800	0.1007 [0.0911, 0.1089]	100%
0.111	900	0.1154 [0.1031, 0.1279]	100%
0.167	1800	0.1708 [0.1594, 0.2204]	100%
0.190	1800	0.1937 [0.1825, 0.2736]	100%
0.200	1800	0.2030 [0.1926, 0.2981]	100%
0.240	900	0.2416 [0.2327, 0.2488]	100%
0.333	900	0.3396 [0.3263, 0.3610]	100%
0.444	1800	0.4488 [0.4381, 0.5059]	100%
0.485	1800	0.4877 [0.4787, 0.5645]	100%
0.500	1800	0.5025 [0.4941, 0.5873]	100%
0.667	900	0.6702 [0.6606, 0.6814]	100%
0.762	1800	0.7634 [0.7551, 0.7828]	100%
0.800	1800	0.8009 [0.7931, 0.8242]	100%
0.889	900	0.8895 [0.8809, 0.8972]	100%
0.941	1800	0.9412 [0.9339, 0.9471]	100%
1.000	900	0.9999 [0.9936, 1.0000]	100%

Table 3: Recovery of the interaction at 24 observations per item and class in 768 or more dimensions, pooled over noise levels, item counts, and widths. *Specified* is the interaction share built into the generated representations and *measured* the median share recovered, with the 2.5th and 97.5th percentiles across runs in brackets; *flagged* is the share of runs whose re-split interval excludes zero.

All three effects read essentially zero when a component is set to zero, at every sample size, and each is flagged at approximately the nominal 5% rate (Table 4).

effect	observations	median	95th pct	flagged
class effect	10	0.00000	0.0081	6.1%
class effect	24	0.00000	0.0034	6.5%
class effect	100	0.00000	0.0009	6.3%
interaction	10	0.00009	0.0644	6.2%
interaction	24	0.00004	0.0309	5.9%
interaction	100	0.00002	0.0083	4.9%
item effect	10	0.00007	0.0568	4.1%
item effect	24	0.00003	0.0255	3.6%
item effect	100	0.00000	0.0064	5.0%

Table 4: What the measure reports when an effect is set to exactly zero, by observations per item and class, pooled over every other axis of the validation grid. *Median* and *95th pct* summarise the measured share of that effect across runs, and *flagged* is the share of runs whose re-split interval excludes zero, which should sit near the nominal 5%.

The false-positive rate varies somewhat with representation width and component, particularly at small widths, but remains below 10% throughout the validation grid except for the class effect at the narrowest width, where it reaches 14% (Table 5). The language-model analyses use $d \geq 768$, where the rates are closer to the nominal level.

width d	item	class	interaction
8	3%	14%	7%
16	7%	3%	5%
32	4%	4%	10%
128	6%	4%	5%
768	2%	8%	3%
1024	3%	5%	3%
2048	0%	8%	8%

Table 5: How often a component set to exactly zero is flagged, by width and by effect.

width d	measured	analytic chance
8	0.17501 [0.06900, 0.40018]	0.25000
16	0.10678 [0.03744, 0.20884]	0.12500
32	0.05609 [0.01864, 0.10454]	0.06250
128	0.01714 [0.00450, 0.02611]	0.02344
768	0.00261 [0.00079, 0.00472]	0.00391
1024	0.00217 [0.00065, 0.00330]	0.00293
2048	0.00096 [0.00031, 0.00165]	0.00146

Table 6: Item-class overlap when the two effects are constructed orthogonal, at 24 observations per item and class. *Measured* is the median overlap across the runs at that width with the 2.5th and 97.5th percentiles in brackets; *chance* is the overlap expected if the component were randomly oriented, r/d for a target subspace of r dimensions in a representation of width d .

With orthogonal item and class effects, the measured overlap tracks the chance level across representation widths (Table 6). When a known fraction of the class direction is placed inside the item subspace, the measured overlap rises above chance and is detected in 89% of runs at $d = 2048$ (Table 7). Thus, the measure recovers both the expected chance overlap under orthogonality and genuine non-orthogonality.

width	d	measured	chance	detected
8	0.1688 [0.0824, 0.3998]	0.12500	9%	
16	0.0951 [0.0414, 0.2026]	0.06250	22%	
32	0.0593 [0.0225, 0.1119]	0.03125	36%	
128	0.0213 [0.0062, 0.0517]	0.00781	63%	
768	0.0065 [0.0011, 0.0298]	0.00260	83%	
1024	0.0056 [0.0008, 0.0331]	0.00098	82%	
2048	0.0036 [0.0004, 0.0290]	0.00049	89%	

Table 7: Measured item-class overlap when a fraction of the class direction is placed inside the item subspace (non-orthogonal), at 24 observations per item and class. *Measured* is the median overlap across the runs at that width with the 2.5th and 97.5th percentiles in brackets; *chance* is the overlap expected if the component were randomly oriented, r/d for a target subspace of r dimensions in a representation of width d ; *detected* is the fraction of runs whose overlap falls outside that null’s 95% interval.

C.1 Variance across contexts

The interaction is a residual, and a residual absorbs whatever noise the terms before it do not. A word’s average representation in one class is taken over the contexts it appears in there, so it contains some amount of sampling error; estimated within a single split, the interaction would inherit that error and grow with the variance across contexts whether or not any interaction was present.

The cross-split dot product eliminates this bias. The two estimates come from disjoint halves of a word’s tokens in that class, so their errors are independent and mean-zero, and the expectation of their inner product is the true squared length with no variance term: the context variance enters the variance of the estimate and not its expectation. This holds for any variance, and for a variance that differs from one word and class to the next.

Table 8 demonstrates this. Holding the planted interaction fixed and raising the variance across contexts to fifty times the variance between words and classes leaves the measured value where it is, and where nothing was planted the false-positive rate stays at the nominal level rather than rising with that variance. If anything the measure drifts slightly low at the largest variance.

specified	noise	runs	measured	flagged
0.00	1x	6300	0.0000	6%
	10x	6300	0.0000	6%
	50x	6300	0.0001	6%
0.10	1x	1800	0.1012	100%
	10x	1800	0.1002	100%
	50x	1800	0.1001	100%
0.20	1x	1800	0.2199	100%
	10x	1800	0.2049	100%
	50x	1800	0.2016	100%

Table 8: Recovery of the interaction across simulations with different amounts of noise, in 768 or more dimensions, pooled over item counts, widths and sample sizes. *Specified* is the interaction share built into the representations and *noise* the variance across contexts per observation as a multiple of the variance between words and classes, so at 1x the two are equal. *Measured* is the median interaction share recovered across runs and *flagged* the share whose re-split interval excludes zero, which should sit at 5% in the rows where nothing was planted.

D Width and Orthogonality

Whether the item and class effects are orthogonal depends on the width of the representation. Figure 9 reports that relationship from the fitted models; this appendix gives the remaining properties of the artificial-language runs.

Under ReLU, overlap stays below chance at wide layers and rises above it as the layer narrows, while the linear condition stays well below chance at every width.

The two summaries that follow are descriptive medians across runs rather than estimates from the fitted models. At low widths, the representations vary substantially across runs, with this variability decreasing as width increases. For example, two runs of the same language differ by 0.70 in the item effect and 0.73 in the class effect at width 8, compared with 0.16 and 0.15 at width 32. This variability reflects the different ways models can distribute information when representational capacity is limited.

Two properties of the language also affect the composition of the item-by-class structure (Table 9). Increasing the effect of context reduces the interaction share, while increasing the item’s share in the language produces a corresponding increase in its share of the representation.

property	level	runs	item	class	interaction
context in the language	under 30%	3600	0.490	0.399	0.028
	30 to 70%	2871	0.538	0.371	0.008
	over 70%	3886	0.578	0.350	0.004
item against class	item < class	3525	0.428	0.480	0.017
	item = class	3427	0.569	0.359	0.015
	item > class	3405	0.644	0.294	0.011

Table 9: How two other properties of the artificial language move the composition of the item-by-class structure, under ReLU, pooled over widths. Rows within a property partition the same runs. Values are medians of the three shares over the runs in the row.

D.1 Component models

Results for our statistical analysis is presented in Tables 10 and 11, and visualized in Figure 7.

component	term	estimate	error	95% CI	P(b > 0)
class	intercept	-0.45	0.04	[-0.53, -0.38]	0.00
	item	-1.47	0.16	[-1.79, -1.15]	0.00
	class	2.01	0.17	[1.67, 2.35]	1.00
	interaction	0.29	0.17	[-0.03, 0.62]	0.96
	width	0.03	0.00	[0.02, 0.04]	1.00
	rank	-0.05	0.03	[-0.10, 0.01]	0.05
	width x rank	-0.01	0.00	[-0.02, -0.01]	0.00
	effect size x width	-0.68	0.02	[-0.71, -0.64]	0.00
	effect size x rank	0.03	0.12	[-0.22, 0.27]	0.58
	effect size x width x rank	-0.03	0.01	[-0.06, -0.00]	0.02
	sd(language)	0.98	0.03	[0.93, 1.04]	1.00
	phi	3.77	0.04	[3.70, 3.85]	1.00
	cutzero	-4.76	0.05	[-4.86, -4.65]	0.00
	cutone	2.22	0.01	[2.20, 2.23]	1.00
interaction	intercept	-3.33	0.03	[-3.39, -3.28]	0.00
	item	-0.17	0.12	[-0.40, 0.05]	0.07
	class	0.12	0.12	[-0.12, 0.35]	0.84
	interaction	2.60	0.12	[2.37, 2.83]	1.00
	width	0.57	0.00	[0.56, 0.58]	1.00
	rank	-0.02	0.02	[-0.05, 0.02]	0.20
	width x rank	0.04	0.00	[0.03, 0.04]	1.00
	effect size x width	-0.19	0.01	[-0.22, -0.17]	0.00
	effect size x rank	0.10	0.08	[-0.06, 0.27]	0.89
	effect size x width x rank	-0.06	0.01	[-0.07, -0.04]	0.00
	sd(language)	0.69	0.02	[0.66, 0.74]	1.00
	phi	26.14	0.36	[25.43, 26.86]	1.00
	cutzero	-5.18	0.03	[-5.23, -5.13]	0.00
	cutone	2.29	0.03	[2.24, 2.35]	1.00
item	intercept	0.22	0.04	[0.15, 0.29]	1.00
	item	1.59	0.16	[1.28, 1.89]	1.00
	class	-1.96	0.17	[-2.29, -1.62]	0.00
	interaction	-0.45	0.16	[-0.76, -0.14]	0.00
	width	-0.09	0.00	[-0.10, -0.09]	0.00
	rank	0.02	0.03	[-0.03, 0.07]	0.82
	width x rank	0.01	0.00	[-0.00, 0.01]	0.97
	effect size x width	-0.12	0.02	[-0.15, -0.09]	0.00
	effect size x rank	-0.14	0.11	[-0.36, 0.08]	0.11
	effect size x width x rank	-0.04	0.01	[-0.07, -0.02]	0.00
	sd(language)	0.95	0.03	[0.90, 1.01]	1.00
	phi	3.38	0.03	[3.31, 3.45]	1.00
	cutzero	-3.84	0.05	[-3.93, -3.74]	0.00
	cutone	2.19	0.01	[2.18, 2.21]	1.00

Table 10: Coefficient values of Equation 3 with error and 95% CIs, ReLU arm.

component	term	estimate	error	95% CI	P(b > 0)
class	intercept	-0.99	0.05	[-1.09, -0.89]	0.00
	item	-3.04	0.22	[-3.48, -2.60]	0.00
	class	4.10	0.20	[3.71, 4.50]	1.00
	interaction	-1.13	0.22	[-1.56, -0.69]	0.00
	width	-0.23	0.00	[-0.24, -0.23]	0.00
	rank	-0.01	0.04	[-0.08, 0.07]	0.44
	width x rank	-0.01	0.00	[-0.01, -0.00]	0.00
	effect size x width	-0.47	0.01	[-0.50, -0.45]	0.00
	effect size x rank	0.03	0.14	[-0.26, 0.30]	0.58
	effect size x width x rank	-0.03	0.01	[-0.05, -0.01]	0.00
	sd(language)	1.41	0.04	[1.34, 1.49]	1.00
	phi	15.06	0.17	[14.72, 15.40]	1.00
	cutzero	-5.66	0.05	[-5.75, -5.57]	0.00
	cutone	2.37	0.01	[2.36, 2.39]	1.00
interaction	intercept	-4.60	0.03	[-4.65, -4.55]	0.00
	item	0.21	0.10	[0.03, 0.41]	0.99
	class	-0.44	0.11	[-0.65, -0.24]	0.00
	interaction	2.07	0.10	[1.88, 2.26]	1.00
	width	0.19	0.00	[0.18, 0.20]	1.00
	rank	-0.08	0.02	[-0.11, -0.05]	0.00
	width x rank	0.03	0.00	[0.02, 0.04]	1.00
	effect size x width	0.36	0.02	[0.33, 0.39]	1.00
	effect size x rank	0.21	0.07	[0.07, 0.34]	1.00
	effect size x width x rank	0.03	0.01	[0.01, 0.05]	1.00
	sd(language)	0.55	0.02	[0.50, 0.59]	1.00
	phi	37.86	0.62	[36.65, 39.08]	1.00
	cutzero	-5.22	0.02	[-5.26, -5.17]	0.00
	cutone	2.26	0.05	[2.16, 2.37]	1.00
item	intercept	0.92	0.05	[0.82, 1.03]	1.00
	item	2.84	0.23	[2.38, 3.30]	1.00
	class	-3.77	0.20	[-4.17, -3.37]	0.00
	interaction	1.23	0.23	[0.77, 1.68]	1.00
	width	0.25	0.00	[0.24, 0.25]	1.00
	rank	-0.00	0.04	[-0.08, 0.07]	0.46
	width x rank	0.01	0.00	[0.00, 0.01]	1.00
	effect size x width	-0.72	0.01	[-0.74, -0.70]	0.00
	effect size x rank	0.01	0.17	[-0.32, 0.33]	0.52
	effect size x width x rank	-0.02	0.01	[-0.04, -0.00]	0.01
	sd(language)	1.47	0.04	[1.39, 1.55]	1.00
	phi	16.93	0.19	[16.56, 17.31]	1.00
	cutzero	-4.67	0.06	[-4.79, -4.55]	0.00
	cutone	2.37	0.01	[2.36, 2.39]	1.00

Table 11: Coefficient values of Equation 3 with error and 95% CIs, linear arm.

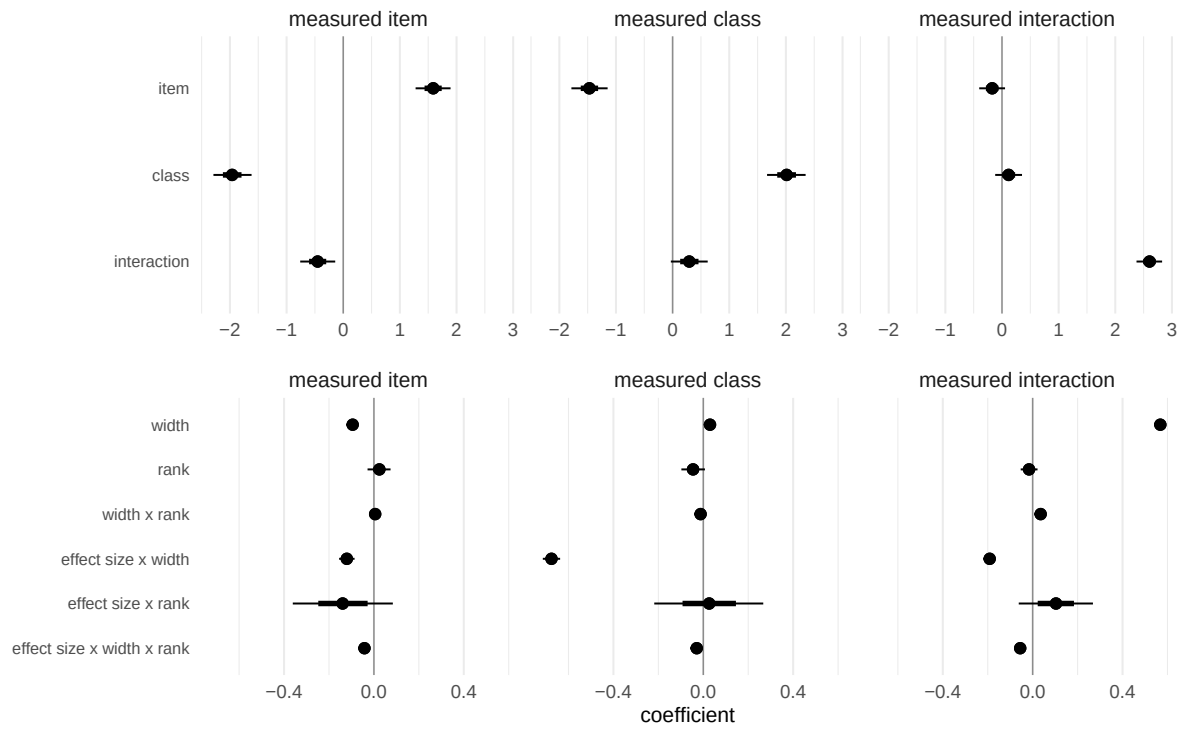


Figure 7: Coefficient estimates from Equation 3, one panel per measured component. The upper block holds the planted effect sizes and the lower the architecture terms, which are on a much smaller scale and so are drawn on their own axis. Points are posterior medians with 95% intervals.

D.2 Overlap models

Table 12 reports every coefficient of Equation 4.

arm	measure	term	estimate	error	95% CI	P(b > 0)
relu	int_margins	intercept	0.49	0.01	[0.47, 0.51]	1.00
	int_margins	item	-1.00	0.05	[-1.10, -0.92]	0.00
	int_margins	class	-0.64	0.05	[-0.74, -0.55]	0.00
	int_margins	interaction	-0.65	0.05	[-0.74, -0.56]	0.00
	int_margins	width	-0.32	0.00	[-0.32, -0.31]	0.00
	int_margins	rank	-0.04	0.01	[-0.05, -0.02]	0.00
	int_margins	width x rank	-0.02	0.00	[-0.03, -0.02]	0.00
	int_margins	sd(language)	0.23	0.01	[0.21, 0.25]	1.00
	int_margins	sigma	0.79	0.00	[0.78, 0.80]	1.00
linear	int_margins	intercept	-0.53	0.02	[-0.56, -0.49]	0.00
	int_margins	item	-1.75	0.08	[-1.92, -1.59]	0.00
	int_margins	class	-0.69	0.09	[-0.86, -0.51]	0.00
	int_margins	interaction	-0.44	0.08	[-0.60, -0.28]	0.00
	int_margins	width	0.10	0.00	[0.10, 0.11]	1.00
	int_margins	rank	-0.05	0.01	[-0.07, -0.02]	0.00
	int_margins	width x rank	-0.02	0.00	[-0.03, -0.02]	0.00
	int_margins	sd(language)	0.49	0.01	[0.46, 0.52]	1.00
	int_margins	sigma	0.73	0.00	[0.72, 0.74]	1.00
relu	item_class	intercept	0.33	0.01	[0.30, 0.36]	1.00
	item_class	item	-0.51	0.07	[-0.65, -0.38]	0.00
	item_class	class	-1.80	0.07	[-1.95, -1.66]	0.00
	item_class	interaction	0.23	0.07	[0.10, 0.36]	1.00
	item_class	width	-0.40	0.00	[-0.41, -0.39]	0.00
	item_class	rank	-0.03	0.01	[-0.05, -0.01]	0.00
	item_class	width x rank	-0.01	0.00	[-0.02, -0.01]	0.00
	item_class	sd(language)	0.37	0.01	[0.34, 0.39]	1.00
	item_class	sigma	0.90	0.00	[0.89, 0.91]	1.00
linear	item_class	intercept	-1.05	0.01	[-1.08, -1.03]	0.00
	item_class	item	-0.58	0.05	[-0.69, -0.47]	0.00
	item_class	class	-0.48	0.06	[-0.59, -0.36]	0.00
	item_class	interaction	0.11	0.05	[0.01, 0.22]	0.98
	item_class	width	-0.00	0.00	[-0.01, 0.00]	0.22
	item_class	rank	-0.05	0.01	[-0.06, -0.03]	0.00
	item_class	width x rank	-0.02	0.00	[-0.02, -0.01]	0.00
	item_class	sd(language)	0.31	0.01	[0.29, 0.33]	1.00
	item_class	sigma	0.64	0.00	[0.64, 0.65]	1.00

Table 12: Every coefficient of Equation 4, with Estimate, Estimated error, and 95% CIs.

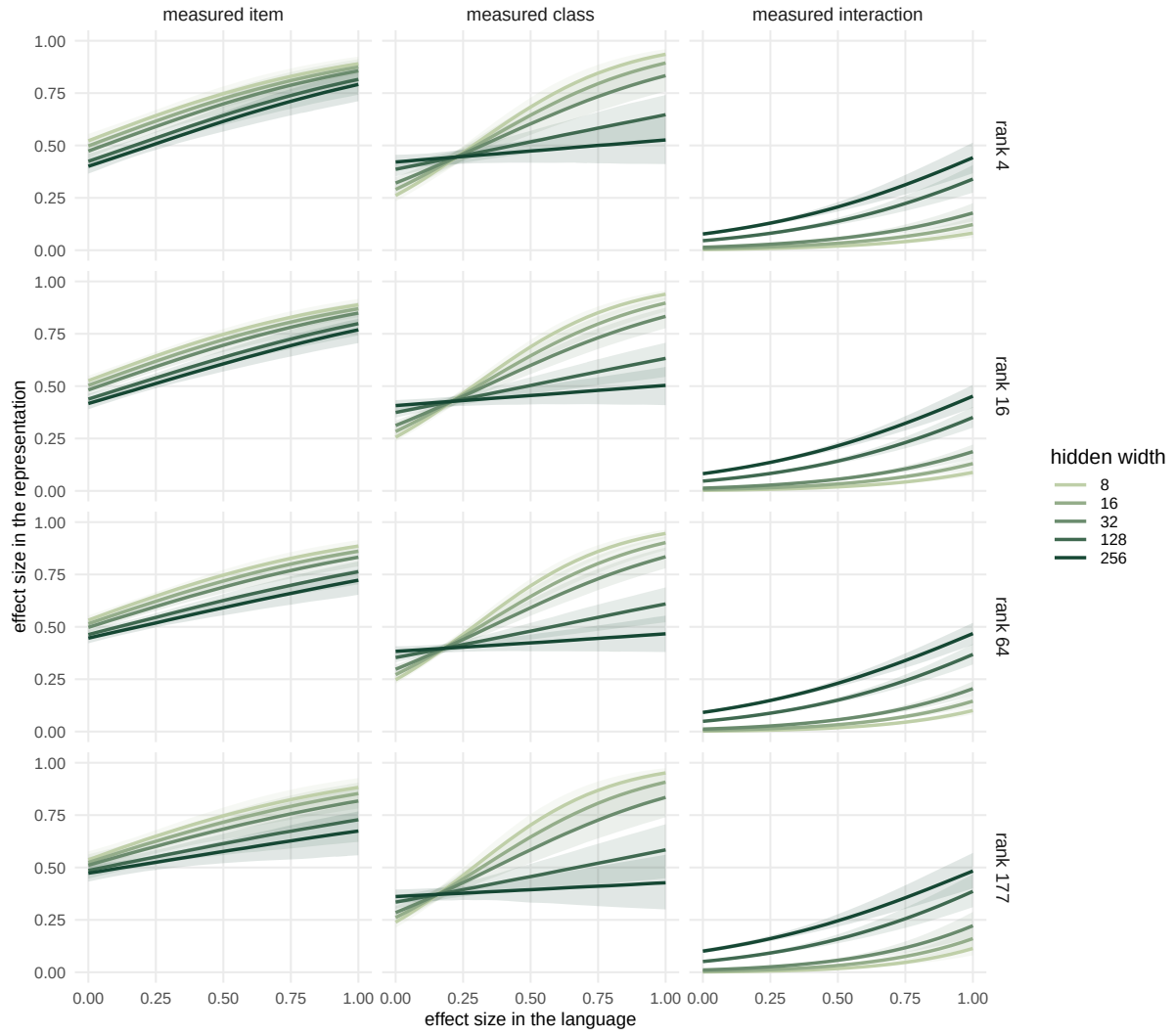


Figure 8: Model results for Equation 3. Columns are the three measured components and lines are hidden widths, with 95% intervals.

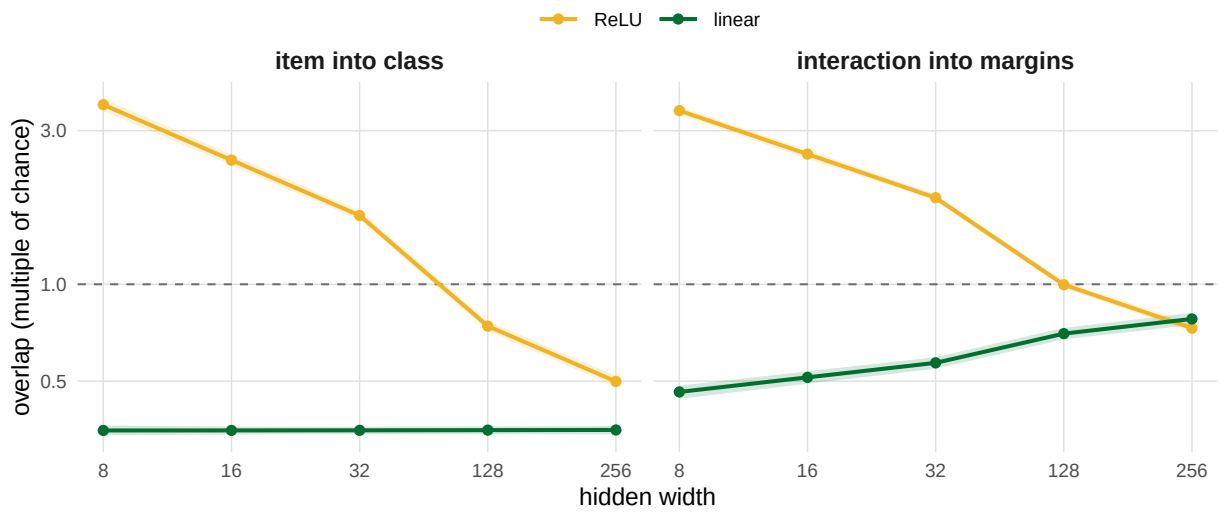


Figure 9: Two overlap measurements as a function of width of the hidden layer. Left: the item effect projected onto the class direction. Right: the interaction projected onto the span of both margins. The dashed line is chance.

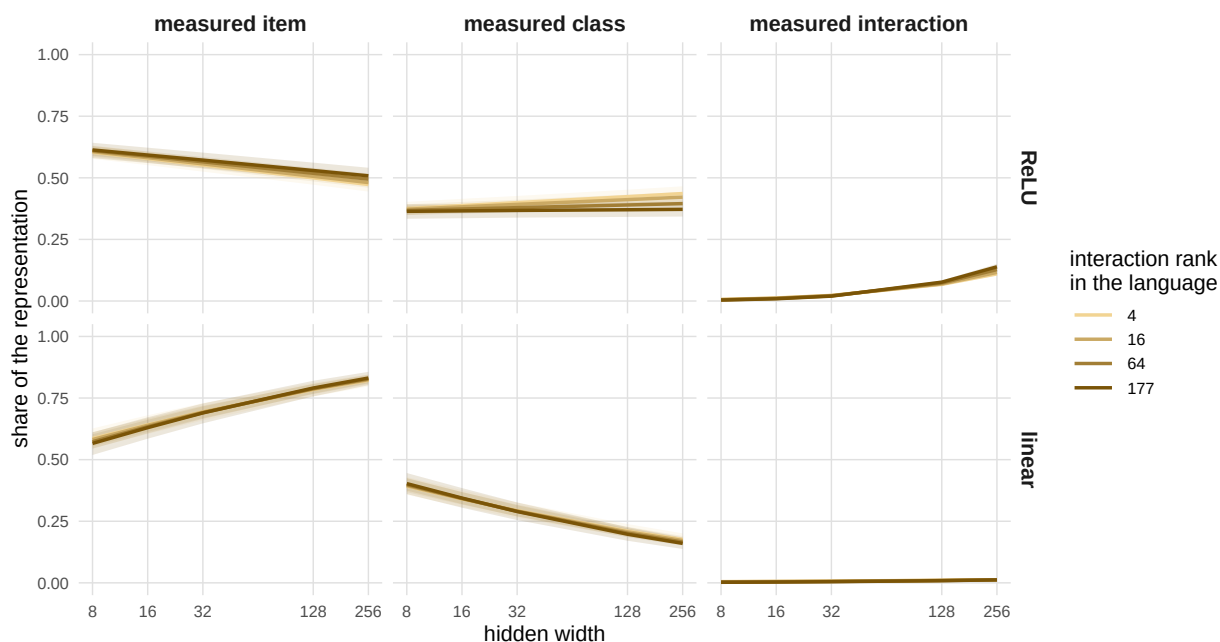


Figure 10: Fraction of the representation by each component (such that they sum to 1), against hidden width, with one line per rank of the interaction planted in the language and 95% intervals.

E Model Details

family	model	params	width	layers	training data	tokens
BabyLM	opt-babylm-125m	125M	768	12	BabyLM (child-available)	~3B
	opt-babylm-350m	350M	1024	24	BabyLM (child-available)	~3B
	opt-babylm-1.3B	1.3B	2048	24	BabyLM (child-available)	~3B
Pythia	pythia-160m	160M	768	12	the Pile	~300B
	pythia-410m	410M	1024	24	the Pile	~300B
	pythia-1.4b	1.4B	2048	24	the Pile	~300B

Table 13: The six models, paired by width. Training-token counts are approximate; the OPT-BabyLM models train for 20 epochs over a 100M-word corpus, which is roughly 150M tokens, so their effective token exposure is about twenty times that.

The six models are listed in Table 13. The two families are matched on width rather than on parameter count, which is what the measure is sensitive to: the chance level for overlap is the class subspace rank over the width, so a comparison across models of different width has to be read against different chance levels. Pairing at 768, 1024, and 2048 removes that from the family comparison.

They differ in two ways that matter elsewhere in the paper. The training corpora differ by more than two orders of magnitude in size, which is the manipulation of interest. They also use different mechanisms for representing the position of each token: Pythia uses rotary embeddings, applied inside attention, while OPT learns absolute position embeddings added at the input. Appendix I uses that difference to rule out absolute position as the source of the interaction.

F Stimulus Counts

class	item types	median tokens
part of speech (noun / verb)	78	18 / 18
grammatical role (subject / object)	49	15 / 19

Table 14: Stimuli for the two linguistic categories, with the two classes of each in parentheses. *Item types* is how many word forms occur in both of them at least ten times; forms occurring in only one are excluded. *Median tokens* is how often a typical form occurs in each of its two classes, in the order they are listed.

G Standardizing the Hidden States

Rogue dimensions could in principle affect what we measure. Table 15 reports part of speech at the deepest layer, measured both with and without the per-dimension rescaling.

model	interaction		class effect		item-class overlap	
	std	raw	std	raw	std	raw
BabyLM 125M	0.095	0.095	0.072	0.074	0.0052	0.0053
BabyLM 350M	0.122	0.122	0.066	0.068	0.0056	0.0061
BabyLM 1.3B	0.104	0.104	0.050	0.051	0.0035	0.0036
Pythia 160M	0.231	0.228	0.078	0.089	0.0058	0.0081
Pythia 410M	0.244	0.242	0.098	0.123	0.0067	0.0195
Pythia 1.4B	0.242	0.238	0.086	0.124	0.0055	0.0234

Table 15: Part of speech at the deepest layer, measured with and without the per-dimension rescaling. Sizes are shares of the item-by-class structure; overlaps are the raw fractions, not multiples of chance.

H Deepest-Layer Measurements

Table 16 gives the results of the overlap measurement at the final layer along with the null.

class	model	item-class overlap	its null	interaction overlap	its null
part of speech	BabyLM 125M	0.0052 (4.2×)	0.0012	0.178 (3.5×)	0.051
	BabyLM 350M	0.0056 (5.9×)	0.0010	0.166 (4.3×)	0.039
	BabyLM 1.3B	0.0035 (7.6×)	0.0005	0.106 (4.8×)	0.022
	Pythia 160M	0.0058 (4.6×)	0.0013	0.231 (4.2×)	0.055
	Pythia 410M	0.0067 (6.9×)	0.0010	0.239 (6.1×)	0.039
	Pythia 1.4B	0.0055 (11.3×)	0.0005	0.209 (9.7×)	0.021
grammatical role	BabyLM 125M	0.0054 (4.0×)	0.0013	0.118 (3.5×)	0.034
	BabyLM 350M	0.0051 (5.5×)	0.0009	0.102 (4.0×)	0.025
	BabyLM 1.3B	0.0033 (6.7×)	0.0005	0.071 (4.7×)	0.015
	Pythia 160M	0.0049 (3.9×)	0.0012	0.179 (5.3×)	0.034
	Pythia 410M	0.0047 (4.9×)	0.0010	0.161 (6.5×)	0.025
	Pythia 1.4B	0.0062 (13.0×)	0.0005	0.141 (11.2×)	0.013

Table 16: Overlap measurement and the null values at the final layer of each model. Each overlap is the fraction of one component lying inside another’s subspace: the item effect inside the class subspace, the interaction inside the span of the two main effects. Each is followed by the overlap divided by its own null, so 4.2x means the overlap is 4.2 times what random orientation gives.

H.1 How much of a token the components account for

The fraction of squared length reported for each measurement demonstrates how much a representation comprises item, class, or interaction components relative to each other. However, it does not say how much of the entire representation comprises that information. In this section, we calculate how much of the overall representation each component takes up (Table 17). We operationalize this by estimating how much of a token’s hidden state each one accounts for as the reduction in squared error against guessing the grand mean. The components are orthogonal, so their contributions accumulate, and the three do not sum to the whole, since most of what varies in a token is the context it appears in.

class	model	word	class effect	interaction
part of speech	BabyLM 125M	36.3	3.1	4.2
	BabyLM 350M	31.2	2.5	4.7
	BabyLM 1.3B	25.8	1.5	3.2
	Pythia 160M	22.4	2.5	7.5
	Pythia 410M	20.3	3.0	7.6
	Pythia 1.4B	18.8	2.4	6.8
grammatical role	BabyLM 125M	38.4	2.1	1.9
	BabyLM 350M	33.0	1.8	2.2
	BabyLM 1.3B	25.9	1.1	1.9
	Pythia 160M	28.0	2.6	3.1
	Pythia 410M	27.5	2.4	2.9
	Pythia 1.4B	24.9	2.4	2.8

Table 17: How much of a token’s hidden state each component accounts for, at the deepest layer, as the percentage reduction in squared error against guessing the grand mean.

I Positional Embeddings Are Not the Source

In theory, one possible architectural explanation for the interaction could be the way in which position is represented within the models. Specifically, the BabyLMs add a *learned absolute position embedding*. The two classes of a linguistic category are not positionally interchangeable in a sentence, since subjects precede objects and nouns and verbs typically occupy different slots, so a representation that varies with position varies with the class. If that is what the decomposition is picking up, the interaction would be an artifact of the position rather than their representation.

In order to rule this out, we re-extracted every OPT-BabyLM model with the positional embeddings zeroed out. Pythia encodes position with rotary embeddings applied inside attention rather than an additive input embedding, so this analysis was run only for the BabyLMs.

We rule out the effect of position by comparing the class effect to the interaction for the models with positions in tact and positions zeroed out. We operationalize this as the class effect divided by the sum of the class effect and the interaction, which we call the *portable fraction*. The three shares sum to one by construction, and zeroing the embeddings drives the item share up sharply, which pushes the class effect and the interaction down together for reasons unrelated to the split between them. Dividing one by their sum holds the item share out of the comparison. If positions were manufacturing the interaction, removing them should mitigate or eliminate the interaction effect and increase the effect of class, so the portable fraction should rise.

model	positions	class	interaction	portable fraction
opt-babylm-125m	zeroed	0.009	0.026	0.258
opt-babylm-125m	intact	0.072	0.095	0.430
opt-babylm-350m	zeroed	0.011	0.043	0.205
opt-babylm-350m	intact	0.066	0.122	0.351
opt-babylm-1.3B	zeroed	0.007	0.032	0.184
opt-babylm-1.3B	intact	0.050	0.104	0.327

Table 18: Trained OPT-BabyLM at the deepest layer, with absolute position embeddings intact and zeroed. Sizes are shares of the item-by-class structure and the portable fraction is the class effect divided by the sum of the class effect and the interaction.

Removing the position embeddings makes these representations *less* separable, not more. The portable fraction falls in every model, from 0.430 to 0.258, from 0.351 to 0.205, and from 0.327 to 0.184 (Table 18). Whatever the absolute position embedding contributes, it is contributing to the part of class structure that *is* carried by a shared direction. Take it away and what remains is more word-specific, not less. The interaction is not an artifact of position.